Magnesium Production in an Aluminum Smelter

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Abstract

Direct MgO reduction in an aluminum smelter could reduce capex and time to market by 60–80% *vs.* a greenfield plant, while producing more magnesium than aluminum at the same current. A reactive liquid tin cathode both keeps the Mg product at the cell bottom and reduces Mg activity to reduce its bath solubility, resulting in >90% current efficiency. Efficient distillation using the G-METS process recovers pure Mg, and tin returns to the electrolysis cell. The present work focuses on chemical, operational, and economic comparisons with aluminum potlines. Similarities with Hall-Héroult cells include operating temperature, carbon anode compatibility, raw material feeding, and exhaust handling. MgO is more effective than Al_2O_3 at scrubbing SO_2, fluorides, and even PFCs from exhaust gases. MgF_2–CaF_2 is less volatile than cryolite, and compatible with zirconia as a solid oxide membrane inert anode or refractory. Thermal insulation and high cathode density can potentially enable very small anode-cathode distance, improving energy efficiency.

Keywords

Electrometallurgy · Extraction and processing · Magnesium · Hall-Héroult · Distillation

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Introduction

Idle aluminum smelters are potentially valuable brownfield sites for industrial development. Six smelters have closed or idled in the past twelve years in the US with a total capacity of 1.25 Mt/a [1]. Some have proposed using their grid connections, often in regions with significant hydropower, for hydrogen or ammonia production. Other electricity-intensive metals such as magnesium (Mg) could also use their grid connections. Either way, the grid connection and transformer/rectifier providing low-voltage high-current DC power would be key assets.

But what if an aluminum smelter could be used intact to produce Mg? That is, what if a new process could reuse not only its high-current DC power supply, but also all of its raw material handling infrastructure, carbon anode plant, potline, off-gas handling, and much of the metal casting facility as well? What if it could directly reduce MgO, avoiding the costly step of $MgCl_2$ dehydration?

To the authors' knowledge, this idea was first proposed by Leland Yerkes over 80 years ago [2]. Put simply, there are two problems with the concept: buoyant low-density Mg product short-circuits the cell; and Mg metal dissolves into the MgO-molten fluoride electrolyte bath, increasing its electronic conductivity, and dramatically reducing current efficiency. To solve both of these problems, Yerkes introduced a reactive cathode, which is a liquid metal forming an alloy with Mg, increasing its density and decreasing its activity and bath solubility. Yerkes proposed using lead for this purpose, but also mentioned tin, zinc, or bismuth as candidates. Electrorefining or distillation would then remove Mg from the cathode alloy.

More recently, a group led by Jungshin Kang proposed a similar process, using the above metals, copper, or silver as the anode [3–11]. This group has focused on MgF_2-LiF as the electrolyte, and on vacuum distillation to remove the Mg product from the cathode alloy. MgF_2-LiF has a relatively low eutectic temperature compared to cryolite, around 720 °C. But

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vacuum distillation as proposed here, and electrorefining as proposed by Yerkes and also Gesing & Das [12–16], are both relatively energy-intensive, using 5–7 kWh/kg Mg product.

To remedy this and further reduce energy use and cost, our group has proposed using gravity-driven multiple effect thermal system (G-METS) distillation to separate Mg from the reactive cathode, and MgF_2–CaF_2 as the electrolyte [17–19]. The former is a variant of METS distillation, which is a nearly 200-year-old concept designed to use condensation energy to power evaporation [20, 21]. Instead of actively pumping liquid, G-METS uses liquid in standpipes to create pressure differences between effects, simplifying operation considerably. This could reduce energy use to as little as 0.5–1.5 kWh/kg Mg product (compared to ΔH_v at 1.42 kWh/kg).

The MgF_2–CaF_2 bath eutectic temperature is 970 °C and close to 50–50 MgF_2–CaF_2. Its use lowers cost and volatility relative to MgF_2-LiF and also uses the CaO impurity in MgO. Incoming CaO reacts with a separate MgF_2 feed to produce MgO and more bath. This is similar to the AlF_3 feed in Hall-Héroult aluminum, which reacts with Na_2O impurity in alumina to create cryolite bath and alumina.

In the past, the magnesium industry has had two choices for reduction technology. The Pidgeon process uses low-cost MgO(CaO) feedstock but requires very high labor and energy inputs and releases extremely high emissions. Alternatively, the efficient chloride electrolysis process requires very costly anhydrous $MgCl_2$ feedstock. This new technology could break that trade-off, using low-cost MgO feedstock in an efficient electrolysis process with relatively low direct emissions. And retrofitting an existing aluminum smelter could reduce capital cost and time to market considerably relative to a new plant, even on a brownfield site. This could potentially make magnesium at lower cost than aluminum, and with lower emissions.

This paper will describe the electrolysis/G-METS distillation process. It will discuss the similarities with Hall-Héroult aluminum which could make it work well in an aluminum smelter. And it discusses differences, risks, challenges, and opportunities for the process moving forward. It is too early to directly compare costs, but this could potentially use less energy in an existing plant to create a higher-value critical material.

Process Description

The new process dissolves MgO in low-volatility MgF_2–CaF_2 molten salt at 1000 °C in aluminum smelter pots with a liquid tin reactive cathode as shown in Fig. 1. A current between a carbon anode or solid oxide membrane (SOM) inert anode and the tin cathode produces Mg-tin alloy and CO_2 or O_2 gas. A new metal distillation technology called gravity-driven multiple effect thermal system (G-METS) separates the Mg product from tin, which returns to the pots.

Although carbon anodes may be useful for rapid deployment, and this process with carbon anodes will likely emit much less CO_2 and other greenhouse gases (GHGs) than the Pidgeon process [19], they are not a suitable long-term solution in a world working toward elimination of GHGs. Fortunately, the MgF_2–CaF_2 bath is more compatible than cryolite with a class of inert anodes as described below.

G-METS reuses the heat of condensation from one "effect" (evaporator-condenser pair) to evaporate metal in the next, as shown in Fig. 2. The weight of liquid metal in a vertical conduit creates a higher pressure in the lower effect, resulting in a higher boiling point, and the hotter condenser below transfers heat to the cooler evaporator above it. Heat thus flows up from effect to effect, reusing it multiple times. Liquid metal flows down from one evaporator to the next, so for example about 23% of the original Mg in the alloy evaporates in the upper effect, 25% in the middle effect, and 27% in the lower effect, leaving the remaining Mg-Sn alloy (concentrate) to return to the pots.

Modeling shows that G-METS for Mg-Sn alloy should use about 1.5 kWh/kg Mg product. Electrolysis uses 10–12 kWh/kg, for a total of 12–14 kWh/kg. This is less than for Al because Mg is divalent and Al trivalent, so a given cell current and power **produces 35% more Mg by mass than Al**.

The process is now at Technology Readiness Level (TRL) 3–4. The Technology Readiness Level estimate is our assessment based on the current scale of the electrolysis and distillation unit operations. Electrolysis current efficiency with carbon anodes is 93.8–97.5% [19], and one batch METS experiment distilled over 5 kg of Mg at >1 kg/hr using <2 kWh/kg Mg product [18].

Similarities with Hall-Héroult

As shown in Fig. 1, the geometry is very similar to Hall-Héroult aluminium cells. Operating temperature around 1000 °C is not much higher than for aluminum production at 950–960 °C. MgO solubility in MgF_2–CaF_2 bath is around

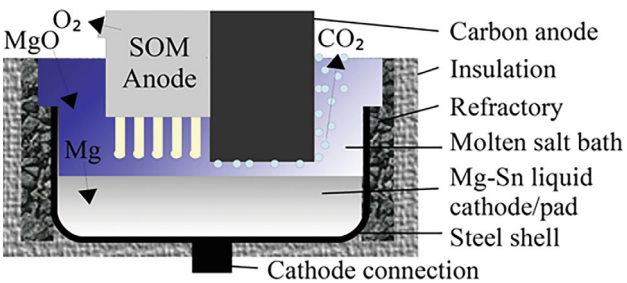


Fig. 1 Cell concept for MgO electrolysis to produce Mg-Sn liquid alloy with carbon and SOM inert anode options

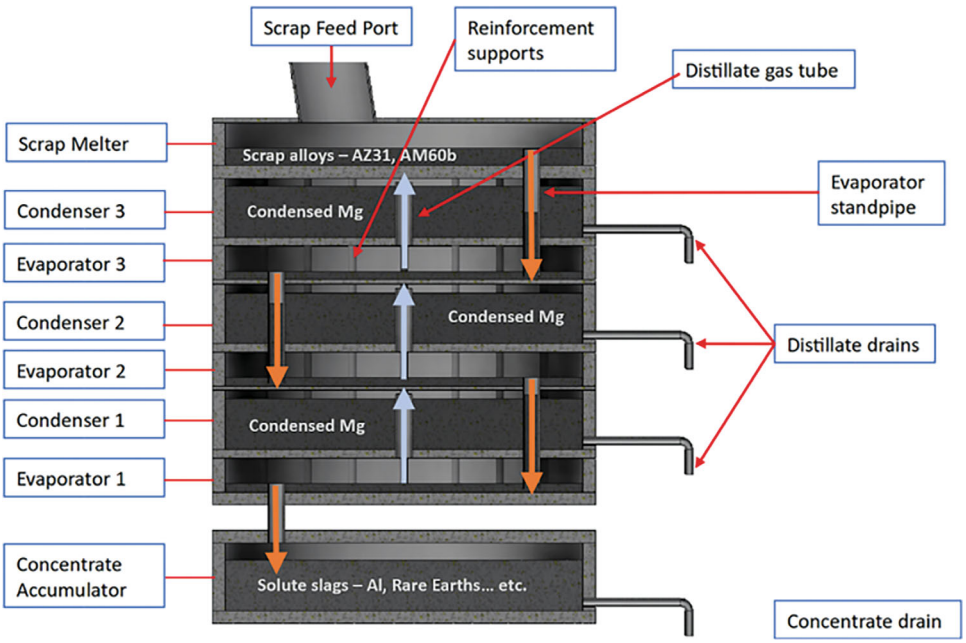


Fig. 2 G-METS distillation schematic

10 wt% [22], which is comparable to alumina solubility in cryolite, and this bath density is only 5–6% denser than cryolite with similar ionic conductivity. Initial trials are likely to use the same graphite pots with iron current collector bars as aluminum production, along with frozen MgF_2–CaF_2 ledge to contain the bath. That said, the low-volatility bath will likely not form crust, so an alternate material will need to act as a radiative heat shield.

The process can also use the existing pot fume hoods and gas handling system in an aluminum smelter. As with Al_2O_3, MgO raw material can scrub any HF from the pot gas, which would form by bath reaction with water vapor. MgO can also absorb SO_2/SO_3 and likely perfluorocarbons as well, leading to lower local acid rain impacts as well as greenhouse gas emissions.

Key Differences, Risks, and Opportunities

While effective scrubbing of SO_2/SO_3 and COS gases by MgO is a process feature, reducing harmful emissions, it can also lead to sulfur build-up in the process. Sulfur enters the process as a carbon anode impurity. If MgO raw material absorbs it as $MgSO_4$ and returns it to the cell, then the sulfur content could increase, with uncertain consequences. MgO may also absorb some CO_2, though not as effectively as CaO would, and this would return carbonates to the cells, where it would volatilize again, resulting in little or no reduction in CO_2 emissions.

If cells operate at the same current density as with aluminum, then they will generate Mg at 35% higher gravimetric rate, and nearly twice the volumetric rate. This will raise the liquid metal height twice as rapidly and likely require tapping twice as frequently. The Hall-Héroult process vacuum siphon tapping crucibles and pipes can likely work well with magnesium, and its mixture with liquid tin reduces both flammability and reactivity with alumina refractory. Each time a pot is tapped, another vessel—likely the same vacuum siphon—will need to bring tin concentrate from the distiller and insert it into the pot. The siphon thus emptied by removing tin can then go to another pot which needs tapping.

In terms of opportunities, although the low density of magnesium leads to higher volumetric rate of accumulation, and although MgF_2–CaF_2 bath density is higher than that of cryolite, the Mg-Sn alloy can nonetheless have a higher density difference *vs.* the bath. This reduces pad swing and can enable lower anode-cathode distance than an aluminum cell.

In addition, yttria-stabilized zirconia is very stable in MgF_2–CaF_2 bath over months of exposure [22]. Zirconia could thus be a potential sidewall refractory material, instead of frozen ledge. This would reduce heat losses dramatically, again allowing a lower anode-cathode distance, as the cells would not need to generate as much heat. This would in turn reduce the cell voltage requirement, and with it, the energy use per tonne of Mg product, relative to Hall-Héroult aluminum.

Steel may also be a potential cathode liner material, as shown in Fig. 1, though only if distillation can remove it to extremely low iron concentration in the product, as some magnesium alloy specifications limit iron to 20 ppm.

As an aside, though the G-METS distiller material of construction will likely be a nickel-free steel, the authors are investigating lining it, for example by using chrome plating, in order to prevent iron dissolution into the magnesium product.

The stability of zirconia in the bath also leads to an inert anode opportunity. First proposed by Marincek [23, 24], this was followed by many others [25–34] due to its benefits. Not only is the anode non-consumable, but oxygen which passes through the solid oxide ion-conductive membrane (SOM) is generally very pure. It also eliminates sulfur and other contaminants brought in by carbon anodes and reduces by 90% or more the volume of gas which must be pumped through the cells. That said, SOM anodes also eliminate gas stirring of the bath, and the large number and relative fragility of SOM anode tubes complicate the cell design and operation considerably.

Finally, retrofitting a plant likely requires changes to the cast house, due mainly to the much higher flammability of magnesium vs. aluminum metal. This is covered extensively elsewhere [35], and will be an issue only if customers require ingots in a format different from those resulting from the distillation process; otherwise this is a downstream issue external to the smelter.

Conclusions

Scaling any new technology is risky, particularly one involving high-temperature electrochemistry. The primary technical risks for the technology described here are in scaling low-cost continuous multiple effect distillation, and unknowns involved in scaling the electrolysis process with its different bath chemistry and properties—particularly sulfur accumulation and the lower volatility of MgF_2–CaF_2. Aluminum production is a finely tuned operation, where everything from feed particle size distribution to macroscopic heat transfer is carefully controlled; this represents a significant change to that operation.

But these risks are much lower than those of completely new processes, and the financial reward to a smelter owner could be great. This is particularly true in the United States where magnesium price is much higher than that of aluminum due to the tariff on Chinese magnesium in place since 2005. And breaking the magnesium industry trade-off between high-cost feedstock in chloride electrolysis and high-cost energy-intensive Pidgeon reduction is a game-changer, let alone the opportunities to reduce heat losses and energy consumption, and use solid oxide membrane inert anodes to eliminate direct emissions.

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References

1. "Aluminum industry in the United States," *Wikipedia*. Sep. 02, 2025. Accessed: Sep. 17, 2025. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Aluminum_industry_in_the_United_States
2. Yerkes LA (1947), "Electrolytic Method for Producing Magnesium," US 2,342,723, Dec. 02, 1947 Accessed: Feb. 20, 2020. [Online]. Available: https://patents.google.com/patent/US2431723A/en
3. Kang J, Lee J-Y, and Lee T-H (2019), "Metal refining method by using liquid metal cathode," KR 10–2004902, Jul. 29, 2019 Accessed: Aug. 26, 2020. [Online]. Available: https://patents.google.com/patent/KR102004920B1/en
4. Kim Y, Yoo J, and Kang J (2020), "Applicability of the Electrochemical Oxygen Sensor for In-Situ Evaluation of MgO Solubility in the MgF2–LiF Molten Salt Electrolysis System," *Metals*, vol. 10, no. 7, p. 906, Jul. 2020, https://doi.org/10.3390/met10070906.
5. Lee T-H, Okabe TH, Lee JY, Kim YM, and Kang J (2020), "Molten Salt Electrolysis of Magnesium Oxide Using a Liquid–Metal Cathode for the Production of Magnesium Metal," *Metall. Mater. Trans. B*, vol. 51, no. 6, pp. 2993–3006, Dec. 2020, https://doi.org/10.1007/s11663-020-01976-9.
6. Lee T-H, Okabe TH, Lee J-Y, Kim YM, and Kang J (2021), "Development of a novel electrolytic process for producing high-purity magnesium metal from magnesium oxide using a liquid tin cathode," *J. Magnes. Alloys*, p. S2213956721000220, Feb. 2021, https://doi.org/10.1016/j.jma.2021.01.004.
7. Lee D-H *et al.* (2021), "Scale-Up Study of Molten Salt Electrolysis using Cu or Ag Cathode and Vacuum Distillation for the Production of High-Purity Mg Metal from MgO," *J. Sustain. Metall.*, vol. 7, no. 3, pp. 883–897, Sep. 2021, https://doi.org/10.1007/s40831-021-00367-x.
8. Kang J *et al.* (2022), "Development of Molten Salt Electrolysis of MgO Using a Metal Cathode and Vacuum Distillation to Produce Ultra-High Purity Mg Metal," in *Rare Metal Technology 2022*, T. Ouchi, G. Azimi, K. Forsberg, H. Kim, S. Alam, N. R. Neelameggham, A. A. Baba, and H. Peng, Eds., in The Minerals, Metals & Materials Series. , Cham: Springer International Publishing, 2022, pp. 309–316. https://doi.org/10.1007/978-3-030-92662-5_29.
9. Kang J, Lee T-H, Jeoung H-J, and Lee JY (2022), "Method for refining metal through molten salt electrolysis-vacuum distillation process of various raw materials containing target metal oxide," KR 10–2402182, May 30, 2022 Accessed: Jun. 19, 2025. [Online]. Available: https://patents.google.com/patent/KR102402182B1/en
10. Jeoung H-J *et al.* (2022), "Use of various MgO resources for high-purity Mg metal production through molten salt electrolysis and vacuum distillation," *J. Magnes. Alloys*, p. S2213956722001712, Aug. 2022, https://doi.org/10.1016/j.jma.2022.07.009.
11. Cho M, Jeoung H-J, Kang J, and Chang K (2023), "Development of Numerical Analysis Model for New Magnesium Electrolysis Process Using COMSOL," *Korean J. Met. Mater.*, vol. 61, no. 8, pp. 625–631, Aug. 2023, https://doi.org/10.3365/KJMM.2023.61.8.625.
12. Gesing AJ, Das SK, and Loutfy RO (2016), "Production of Magnesium and Aluminum-Magnesium Alloys from Recycled Secondary Aluminum Scrap Melts," *JOM*, vol. 68, no. 2, pp. 585–592, Feb. 2016, https://doi.org/10.1007/s11837-015-1720-1.
13. Gesing AJ, Das SK, and Loutfy RO (2016), "Production of Magnesium and Aluminum-Magnesium Alloys from Recycled Secondary Aluminum Scrap Melts," in *Rewas 2016: Towards Materials Resource Sustainability*, R. E. Kirchain, B. Blanpain, C. Meskers, E. Olivetti, D. Apelian, J. Howarter, A. Kvithyld, B. Mishra, N. R. Neelameggham, and J. Spangenberger, Eds., Hoboken, NJ, USA: John Wiley & Sons, Inc., 2016, pp. 253–254. https://doi.org/10.1002/9781119275039.ch38.

14. Gesing AJ and Das SK (2017), "Use of Thermodynamic Modeling for Selection of Electrolyte for Electrorefining of Magnesium from Aluminum Alloy Melts," *Metall. Mater. Trans. B*, vol. 48, no. 1, pp. 132–145, Feb. 2017, https://doi.org/10.1007/s11663-016-0724-8.
15. Gesing AJ, Das S, and Gesing MA (2025), "Electrorefining of Magnesium from Scrap Metal Aluminum or Magnesium Alloys," US 10,017,867, Jul. 10, 2018 Accessed: Sep. 15, 2025. [Online]. Available: https://patents.google.com/patent/US10017867B2/en
16. Gesing AJ and Das SK (2020), "Electrorefining of Magnesium from Scrap Metal Aluminum or Magnesium Alloys," US 10,557,207, Feb. 11, 2020 Accessed: Jan. 15, 2023. [Online]. Available: https://patents.google.com/patent/US10017867B2/en
17. Powell AC, Dussault DM, Earlam MR, Tajima A, and Raymes C (2023), "Method and Apparatus for Efficient Metal Distillation and Related Primary Production Process," US 11,773,500, Oct. 03, 2023 Accessed: Nov. 05, 2023. [Online]. Available: https://patents.google.com/patent/US11773500B2
18. Rutherford M, Telgerafchi AE, Espinosa G, Powell AC, and Dussaul DT (2021), "Low-Cost Magnesium Primary Production Using Gravity-Driven Multiple Effect Thermal System (G-METS) Distillation," in *Magnesium 2021*, Orlando, Florida: TMS, Mar. 2021, pp. 139–144. https://doi.org/10.1007/978-3-030-65528-0_21.
19. Telgerafchi AE *et al.* (2023), "Magnesium production by molten salt electrolysis with liquid tin cathode and multiple effect distillation," *Front. Chem.*, vol. 11, p. 1192202, Jul. 2023, https://doi.org/10.3389/fchem.2023.1192202.
20. Rillieux N (1843), "Improvement in Sugar-Works," US 3,237, Aug. 26, 1843 Accessed: Sep. 15, 2025. [Online]. Available: https://patents.google.com/patent/US3237A/en
21. Weldon N (2025), "The Free Man of Color Who Revolutionized the Sugar Industry," The Historic New Orleans Collection. Accessed: Sep. 15, 2025. [Online]. Available: https://hnoc.org/publishing/first-draft/free-man-color-whose-invention-revolutionized-sugar-industry
22. Gratz ES, Milshtein JD, and Pal UB (2013), "Determining Yttria-Stabilized Zirconia (YSZ) Stability in Molten Oxy-Fluoride Flux for the Production of Magnesium with the SOM Process," *J. Am. Ceram. Soc.*, vol. 96, no. 10, pp. 3279–3285, 2013, https://doi.org/10.1111/jace.12449.
23. Marincek B (1971), "Electrolytic Cell," US 3,562,135, Feb. 09, 1971 Accessed: Sep. 17, 2025. [Online]. Available: https://patents.google.com/patent/US3562135A/en
24. Marincek B (1972), "Electrolytic Process," US 3,692,645, Sep. 19, 1972 Accessed: Sep. 17, 2025. [Online]. Available: https://patents.google.com/patent/US3692645A/en
25. Schmidt-Hatting W and Huwyler S (1971), "Electrolytic Cell Apparatus," US 3,578,580, May 11, 1971 Accessed: Sep. 16, 2025. [Online]. Available: https://patents.google.com/patent/US3578580A/en
26. Ginsberg H and Wilkening S (1975), "Verfahren und Vorrichtung zur schmelzflußelektrolytischen Reduktion von Metalloxiden zu ihren Metallen," German Patent 19 48 462, Jan. 22, 1975 [Online]. Available: https://patents.google.com/patent/DE1948462B2/de
27. Minck R (1978), "Method and apparatus for separating a metal from a salt thereof," US 4,108,743, Aug. 22, 1978 [Online]. Available: https://patents.google.com/patent/US4108743A/en
28. Rapp RA (1999), "Method and Apparatus Featuring a Non-Consumable Anode for the Electrowinning of Aluminum," US 5,942,097, Aug. 24, 1999 Accessed: Sep. 16, 2025. [Online]. Available: https://patents.google.com/patent/US5942097A/en
29. LaCamera AF and Ray SP (2001), "Electrolysis in a Cell Having a Solid Oxide Ion Conductor," US 6,187,168, Feb. 13, 2001 Accessed: Sep. 16, 2025. [Online]. Available: https://patents.google.com/patent/US6187168B1/en
30. Krishnan A, Pal UB, and Lu XG (2005), "Solid oxide membrane process for magnesium production directly from magnesium oxide," *Metall. Mater. Trans. B*, vol. 36, no. 4, pp. 463–473, Aug. 2005, https://doi.org/10.1007/s11663-005-0037-9.
31. Krishnan A, Lu XG, and Pal UB (2005), "Solid Oxide Membrane (SOM) technology for environmentally sound production of tantalum metal and alloys from their oxide sources," *Scand. J. Metall.*, vol. 34, no. 5, pp. 293–301, Oct. 2005, https://doi.org/10.1111/j.1600-0692.2005.00749.x.
32. Jiang Y, Xu J, Guan X, Pal UB, and Basu SN (2013), "Production of Silicon by Solid Oxide Membrane-Based Electrolysis Process," *MRS Proc.*, vol. 1493, pp. 231–235, 2013, https://doi.org/10.1557/opl.2013.33.
33. Jiang Y, Xu J, Lo B, Pal UB, and Basu SN (2013), "Production of Silicon from Silica: Solid-Oxide-Membrane Based Electrolysis Process," in *EPD Congress 2013*, John Wiley & Sons, Ltd, 2013, pp. 173–183. https://doi.org/10.1002/9781118658468.ch20.
34. Powell A, EarlamM, and Barriga S (2014), "Pure Oxygen Anodes^{TM} for Low- or Zero-Carbon Energy Efficient Metal Oxide Reduction," in *Energy Technology 2014*, C. Wang, J. de Bakker, C. K. Belt, A. Jha, N. R. Neelameggham, S. Pati, L. H. Prentice, G. Tranell, and K. S. Brinkman, Eds., Hoboken, NJ, USA: John Wiley & Sons, Inc., 2014, pp. 125–134. https://doi.org/10.1002/9781118888735.ch16.
35. Baker PW' and McGlade PT (2016), "Magnesium Direct Chill Casting: A Comparison with Aluminium," in *Essential Readings in Light Metals,* vol. 4, pp. 542–549, 2016. https://doi.org/10.1007/978-3-319-48228-6_65.